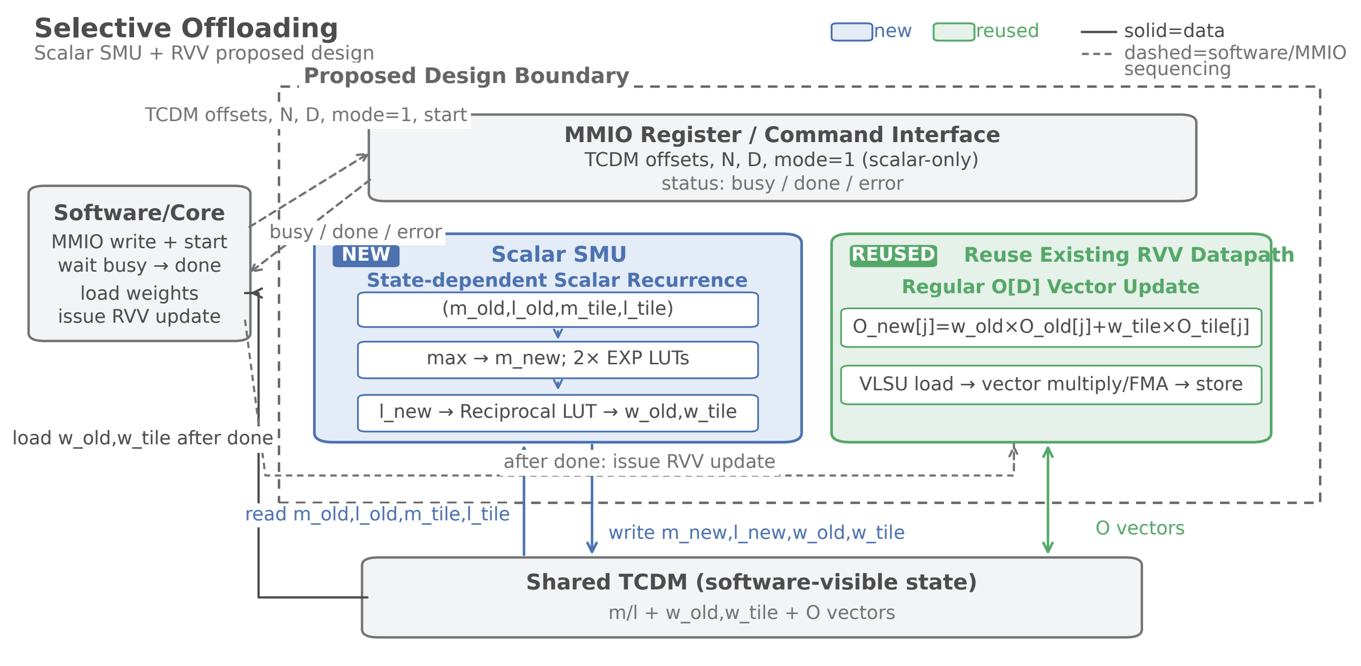

Selective Offloading
Scalar SMU + RVV proposed design
new
reused
solid=data
dashed=software/MMIO sequencing
Proposed Design Boundary
TCDM offsets, N, D, mode=1, start
MMIO Register / Command Interface
TCDM offsets, N, D, mode=1 (scalar-only)
status: busy / done / error
Software/Core
MMIO write + start
wait busy → done
load weights
issue RVV update
busy / done / error
NEW
Scalar SMU
State-dependent Scalar Recurrence
(m_old,l_old,m_tile,l_tile)
max → m_new; 2× EXP LUTs
l_new → Reciprocal LUT → w_old,w_tile
REUSED
Reuse Existing RVV Datapath
Regular O[D] Vector Update
O_new[j]=w_old×O_old[j]+w_tile×O_tile[j]
VLSU load → vector multiply/FMA → store
load w_old,w_tile after done
after done: issue RVV update
read m_old,l_old,m_tile,l_tile
write m_new,l_new,w_old,w_tile
O vectors
Shared TCDM (software-visible state)
m/l + w_old,w_tile + O vectors